

Cloudflare R2 archive

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Cloudflare R2 stores selected experiment scratch that is expensive to reproduce, especially trained checkpoint banks under `temp/experiments/`. Git remains the record for source code and published artifacts. R2 is a manual backup, not a live training filesystem.

The supported interface is `experiments/helpers/archive.py`. It stores snapshots at:

```
r2:pinglab/archive/<experiment-slug>/<producing-git-sha>/
```

Access and authentication

The Mac currently has the `r2:` rclone remote. Hetzner does not. Credentials consist of an Access Key ID, Secret Access Key, and the EU R2 endpoint. Never commit or paste their values.

To configure a new machine, create an **Object Read & Write** token scoped to the `pinglab` bucket in the Cloudflare dashboard. Then configure rclone interactively:

```
rclone config
```

Use these choices:

1. Remote name: `r2`.
2. Storage: `s3`.
3. Provider: `Cloudflare`.
4. Enter the Access Key ID and Secret Access Key manually.
5. Endpoint: the EU endpoint shown by Cloudflare.
6. ACL: `private`.

Prefer a separate bucket-scoped token for each machine. See Cloudflare's token guide and rclone guide.

Verify access without writing:

```
rclone listremotes
rclone config redacted r2
rclone lsd r2:
rclone lsf --dirs-only r2:pinglab/archive
```

Do not print the unredacted rclone configuration.

Archive a run

Check the local scratch, then archive it:

```
du -sh temp/experiments/exp022
uv run python experiments/helpers/archive.py archive exp022
```

The helper determines the producing Git commit, copies the tree, uploads `MANIFEST.json`, and verifies it with `rclone check`. It uses `copy`, never `sync`, so a partial local tree cannot delete existing remote objects.

List snapshots

```
uv run python experiments/helpers/archive.py list exp022

rclone tree r2:pinglab/archive/exp022 --max-depth 2
rclone size r2:pinglab/archive/exp022
rclone cat r2:pinglab/archive/exp022/<sha>/MANIFEST.json
```

Restore a snapshot

Restore the latest snapshot:

```
uv run python experiments/helpers/archive.py restore exp022
```

Or restore a specific producing commit:

```
uv run python experiments/helpers/archive.py restore exp022 <sha>
```

Restore copies into `temp/experiments/exp022/` without deleting unrelated local files. Validate configuration files and load representative checkpoints before using the restored bank.

Safety rules

- Never run `rclone sync` against an archive prefix.
- Archive only completed runs worth preserving.
- Do not use R2 as mutable training storage.
- Keep snapshots from different producing commits separate.
- Revoke and replace a token immediately if it may have leaked.
- Do not confuse R2 credentials with `RUNPOD_S3_ACCESS_KEY_ID` and `RUNPOD_S3_SECRET_ACCESS_KEY`; those access RunPod storage.

The helper deliberately has no delete command. Remote deletion should remain a separate, explicit administrative action.